Module 4: Supervised Learning
Submodule: 4.6 Gaussian Discriminant Analysis
Lecture: Generative Learning Algorithms

XCS229 Machine Learning

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Contents

I. Introduction

In this lecture, we will discuss **generative learning algorithms**. We will define what it means by generative learning algorithms and cover two types:

- **Gaussian Discriminative Analysis (GDA)**
- **Naive Bayes**

II. Discriminative Learning Algorithms

Discriminative learning algorithms model the conditional relationship of y given x. Therefore, the main focus is:

$$p(y \mid x)$$

In the previous lectures, we modeled y as a function of x. For example, we can express this as:

$$y \mid x; \theta = \text{Exponential family}(\eta), \qquad \eta = \theta^T x.$$

where η is a linear function of x. This is why we refer to these as discriminative learning algorithms. For example, you can say this is a Gaussian distribution, with mean η. And that's just the standard linear regression.

III. Generative Learning Algorithms

Generative learning algorithms aim to model or parameterize the joint distribution $p(x, y)$. This is expressed mathematically as:

$$p(x, y) = p(x \mid y) \cdot p(y)$$

Here:

- x is the inputs
- y a label or some kind of class
- $p(y)$ is the distribution of the labels (also called a prior for the class or the label)
- $p(x \mid y)$ is the distribution of the input given the label

Note that x and y are not symmetric; typically, y represents the label (outcome), whereas x is the input features. For example, if y is the price of a house, then x could include features such as square footage, lot size, etc.

Here, $p(y)$ is sometimes referred to as the prior distribution for the label or class. This reflects what we believe about the distribution of classes before observing any data. In classification problems, this could be:

$$\begin{cases} y = 1 & \text{(positive sentiment)} \\ y = 0 & \text{(negative sentiment)} \end{cases}$$

This indicates the distribution over two labels in our dataset.

IV. Learning the Distributions

After modeling and parameterizing both distributions, we aim to learn them:

$$p(x \mid y) \quad \text{and} \quad p(y)$$

To solve the classification problem, we still want to compute $p(y \mid x)$. We will use **Bayes' rule**:

$$p(y \mid x) = \frac{p(x \mid y) \cdot p(y)}{p(x)}.$$

To understand $p(x)$, we can compute it using the law of total probability:

$$p(x) = \sum_{y'} p(x \mid y') \cdot p(y').$$

We denote y' as the variable from the summation. Thus,

$$p(y \mid x) = \frac{p(x \mid y) \cdot p(y)}{p(x)} = \frac{p(x \mid y) \cdot p(y)}{\sum_{y'} p(x \mid y') \cdot p(y')}.$$

Using Bayes' theorem:

- You will know $p(x \mid y)$ and $p(y)$.
- The denominator $p(x)$ can be computed as shown, although in many cases you may not need to compute it directly.

After knowing these quantities, $p(y \mid x)$ can be found using Bayes' rule.

While discriminative learning algorithms require only the conditional distribution $p(y \mid x)$ to be parameterized by some θ, generative algorithms require a full description of the joint distribution. If the distributions belong to the exponential family, there are several advantages.

V. Summary

- **Discriminative Learning:** focuses on modeling $p(y \mid x)$, the conditional probability of the label given the input.
- **Generative Learning:** focuses on modeling the joint distribution $p(x, y) = p(x \mid y) \cdot p(y)$.
- **Bayes' rule:** is applied to compute $p(y \mid x)$ from $p(x \mid y)$ and $p(y)$.
- **Gaussian Discriminant Analysis (GDA) and Naive Bayes:** are examples of generative models.
- **Generative models:** are beneficial when the distributions follow the exponential family due to their flexibility and richness in modeling data.